

COLD vs CCDC: break-date comparison on the Holiday Farm Fire

Change-dating comparison between **CCDC** (`ee.Algorithms.TemporalSegmentation.Ccdc` , run in Google Earth Engine) and **COLD** (Zhu et al. 2020, run locally via the `pyxccd` library) on the 2020 Holiday Farm Fire pilot. Both algorithms are dated against the known ignition date, **2020-09-08**. Each is run at two confirmation-threshold settings so the comparison is fair (CCDC's `minObservations` and COLD's `conse` are the same lever: the number of consecutive residual-exceeding observations required to confirm a break).

Sample

50 points drawn from the classifier's own Trees to non-Trees change map (same seed as the pilot):

- **30 random** change points (any severity, any transition)
- **10 stratified Trees to Water** points
- **10 stratified Trees to Developed** points

Water and Developed transitions are largely classifier noise rather than real fire change, and are included deliberately as a false-positive check.

Methods compared

Label	Algorithm	Confirmation setting	Where it runs
CCDC minObs=6	CCDC	<code>minObservations=6</code> (default)	GEE (server-side)
CCDC minObs=4	CCDC	<code>minObservations=4</code> (pilot's tuned value)	GEE (server-side)
COLD conse=6	COLD (pyxccd)	<code>conse=6</code> (default)	local (isolated venv)
COLD conse=4	COLD (pyxccd)	<code>conse=4</code>	local (isolated venv)

All four use `p_cg / chiSquareProbability = 0.99` , `lambda = 20` , and the same Landsat 7/8/9 Collection-2 surface-reflectance observations over 2017-01 to 2026-08.

Accuracy

Break date measured against the 2020-09-08 ignition, over the 50 points.

Method	Breaks found	On-target (within 25 days)	Spurious pre-fire	Median distance from ignition
CCDC minObs=6	36 / 50	15	1	176 days
CCDC minObs=4	41 / 50	22	2	8 days
COLD conse=6	42 / 50	23	1	8 days
COLD conse=4	45 / 50	30	1	1 day

- **On-target** = break within +/-25 days of the ignition.
- **Spurious pre-fire** = break more than 40 days *before* the ignition (a likely false positive).
- **Median distance** = median of the absolute distance in days from the ignition, over points that had a break. Because most points sit inside the fire, closer to zero is better; the metric does fold in genuine non-fire changes (some

Developed/Water points), so read it as a relative comparison across methods on identical points, not an absolute accuracy.

COLD dates the fire markedly closer than CCDC at matched settings, and with no increase in spurious pre-fire breaks. At the tuned setting, COLD conse=4 puts 30 of 50 points within 25 days of the ignition (vs 22 for CCDC minObs=4) and has a median distance of 1 day. Even at its default conse=6, COLD matches CCDC's *tuned* minObs=4 result.

Timing

Method	Total (50 points)	Per point	Notes
CCDC minObs=6	321.0 s	6.42 s	GEE server-side compute + one <code>getInfo</code> per point
CCDC minObs=4	281.5 s	5.63 s	same
COLD band extraction	322.3 s	6.45 s	one-time GEE <code>getRegion</code> per point (shared by both COLD runs)
COLD conse=6	0.343 s	0.007 s	local compute only
COLD conse=4	0.342 s	0.007 s	local compute only

Reading the timings. The two algorithms spend their time in completely different places. CCDC's cost is dominated by GEE round-trips: every configuration re-run pays the full ~5-6 s per point again (the two CCDC runs cost ~600 s combined). COLD's cost is almost entirely the one-time band extraction from GEE (~6.5 s per point); once the observations are on disk, the COLD algorithm itself runs in ****~7 milliseconds per point****, roughly a thousand times faster than a CCDC round-trip. So re-running or re-tuning COLD (e.g. sweeping `conse`) is effectively free, whereas each CCDC re-tune is another full GEE pass. End-to-end for a single setting the two are comparable (~6.5 s/point); COLD pulls ahead sharply the moment you iterate.

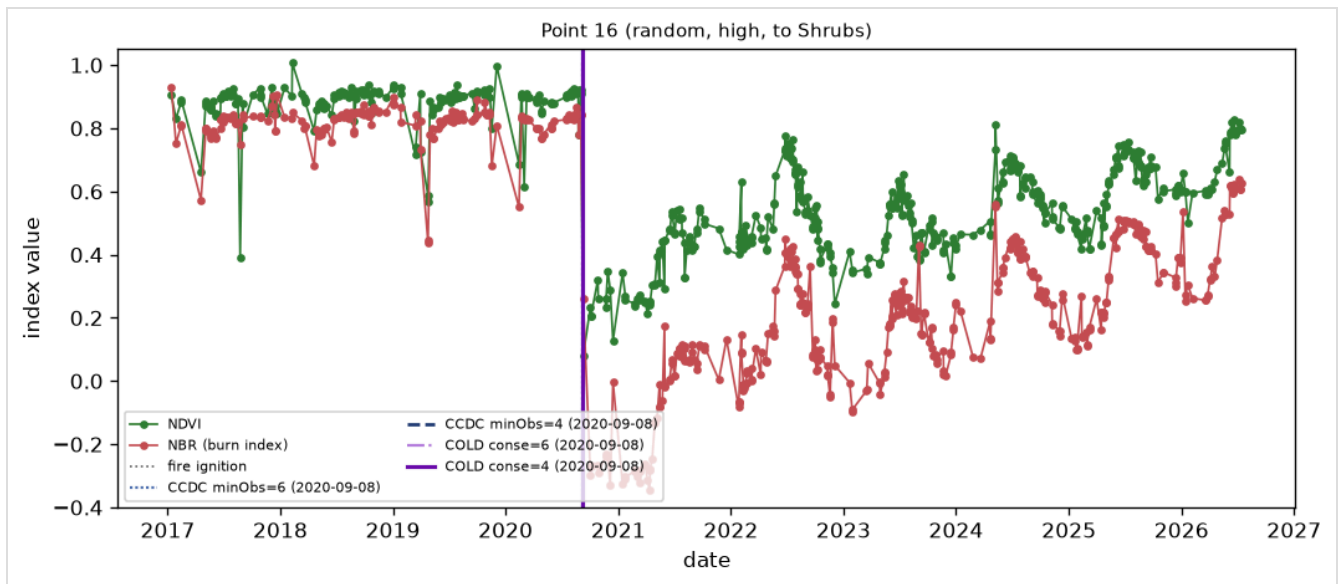
This is a 50-point test, so the absolute numbers are small and local (about zero dollars). At wall-to-wall scale COLD's economics invert: it must move Landsat stacks out of GEE (ARD tiles, egress, a VM), which is where its real cost lives.

Select time series

Each panel shows the point's clear-observation NDVI (green) and NBR / burn index (red), the fire ignition (black dotted), and all four break dates: CCDC minObs=6 (light blue dotted), CCDC minObs=4 (dark blue dashed), COLD conse=6 (light purple dash-dot), COLD conse=4 (solid purple).

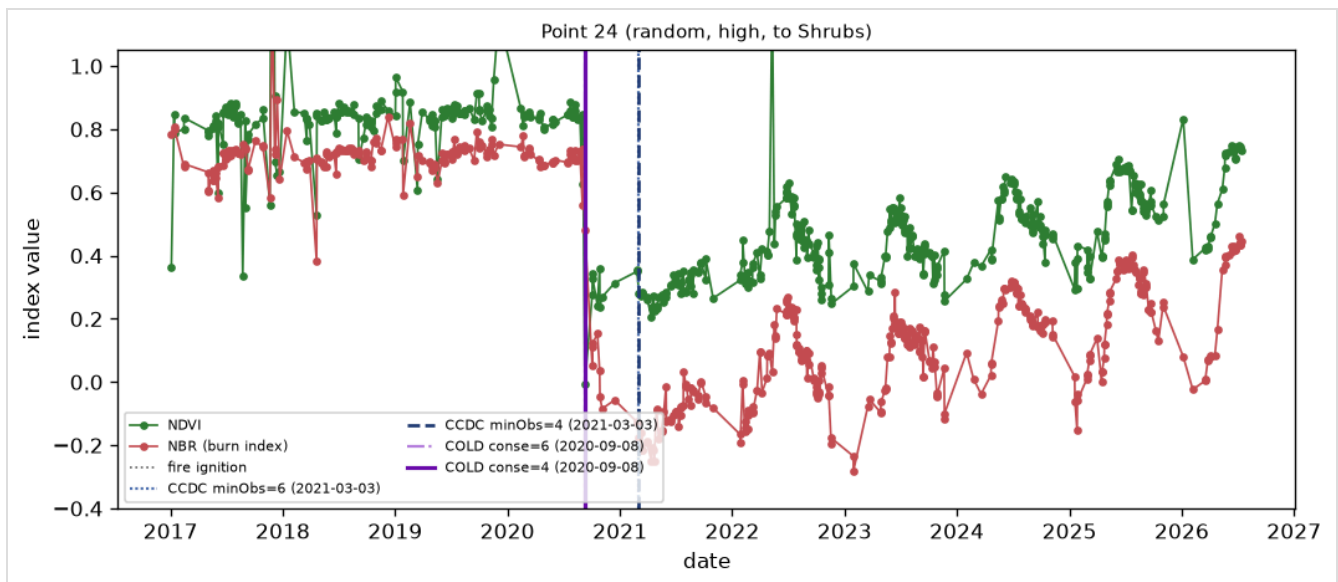
1. Agreement on a clean fire pixel (point 16, high severity)

All four methods land on 2020-09-08, exactly on the NDVI/NBR crash. On unambiguous stand-replacing pixels the two algorithms agree.



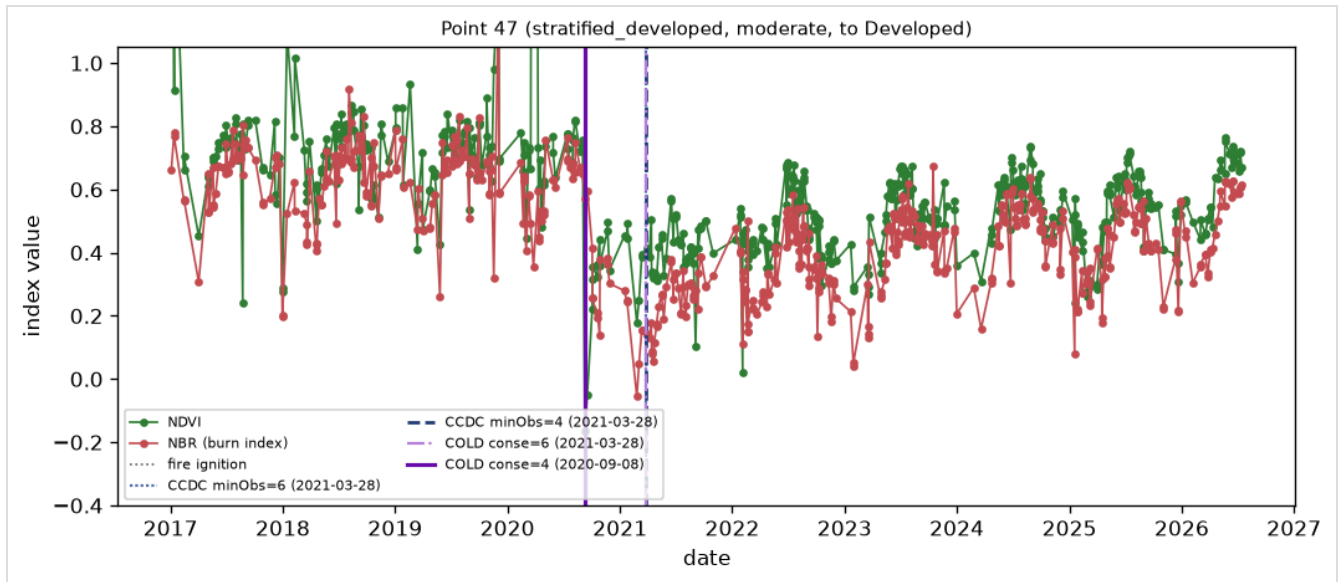
2. COLD fixes a CCDC lag (point 24, high severity)

The crash is unmistakably at September 2020. Both COLD settings date it to 2020-09-08; both CCDC settings lag ~6 months to 2021-03-03, landing deep in the post-fire trough. This is a genuine high-severity fire pixel that CCDC mis-dates and COLD gets right.



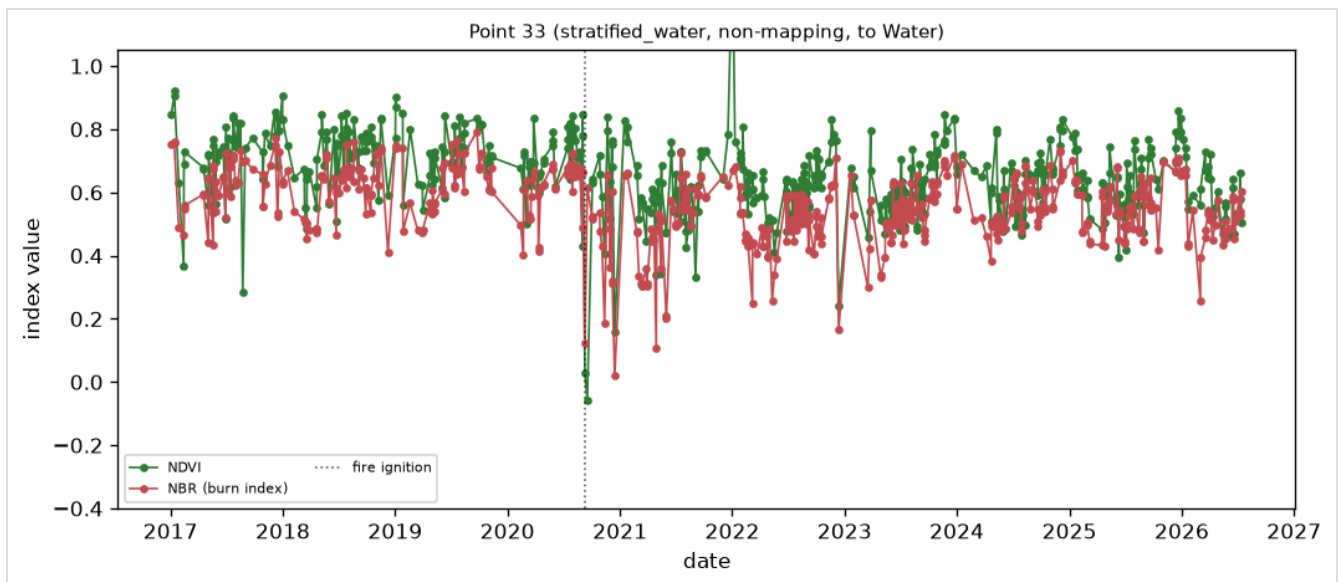
3. The confirmation-threshold lever affects COLD too (point 47, Developed)

COLD conse=4 recovers the September 2020 date, while COLD conse=6 and CCDC minObs=4 both lag to 2021-03-28. The `conse / minObs` setting matters for both algorithms; COLD at conse=4 simply recovers more of the true fire timing.



4. Classifier noise, correctly left undated (point 33, Water)

A stratified Trees to Water point with no real discontinuity. No method reports a break, which is the correct null result: absence of a break is the signal that the classifier's "change" here is noise, not a real event.



The full 50-page per-point comparison is in [CCDC Outputs/cold vs ccdc timeseries n30 water10 dev10.pdf](#), and the per-point break table is in [CCDC Outputs/cold vs ccdc breaks n30 water10 dev10.csv](#).

Caveats

- **One fire, one disturbance type.** This tests abrupt disturbance dating, which is COLD's designed strength. It does not establish that COLD is better for gradual change.
- **Not a masking-controlled comparison.** COLD used a cfmask decoded from Landsat QA_PIXEL ; GEE-CCDC used its own internal masking. The input scenes are the same, so this is a fair "as-run" comparison, not a pure algorithm-only diff.
- **Off-GEE dependency.** COLD is not available in Earth Engine; it runs through `pyxccd` in an isolated environment (it requires numpy 2.x, which conflicts with the project's `gee env`). For a point test that is trivial and free; at wall-to-wall scale it means moving data out of GEE.

- **Metric mixes strata.** The accuracy metrics treat "near 2020-09-08" as correct, which is right for fire pixels but not necessarily for the Developed/Water stratified points. All methods run on identical points, so the comparison is valid; the absolute numbers should not be read as a fire-only accuracy.

Reproducibility

- CCDC: `Change_Detection.run_ccdc(..., min_observations=6|4)` over the 50 points.
- COLD: `pyxccd 1.0.3, cold_detect(..., conse=6|4, p_cg=0.99, lam=20, b_c2=True)` , fed Landsat C2 L2 surface reflectance scaled to [0, 10000], thermal as Kelvin x 10, and a cfmask QA decoded from `QA_PIXEL` . Validated by confirming a known high-severity pixel dates to 2020-09-08 before trusting the batch.